

BUKU PANDUAN MASTER ENSIKLOPEDIS PLATFORM WISUDA v2.0

Spesifikasi Lengkap Arsitektur, Skema Database, Workflow SOP, Integrasi Google Drive,
Media Handling, API Endpoints & Troubleshooting

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Spesifikasi Lengkap Arsitektur, Skema Database, Workflow SOP, Integrasi Drive, Media Handling, API Endpoints, Matriks Tombol UI, & Troubleshooting

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Bab 1: Gambaran Umum, Arsitektur Sistem & Arsitektur Kode

1.1 Pendahuluan

Platform Wisuda v2.0 adalah sistem manajemen operasional foto wisuda berbasis web yang terintegrasi secara utuh untuk mengelola siklus operasional bisnis fotografi wisuda dari prospek awal (*inquiry*), konfirmasi booking client, penugasan fotografer freelance, pengolahan media pasca produksi di Google Drive, hingga pelunasan pembayaran dan payroll.

1.2 Struktur Folder Proyek

```
/Wisuda/
  ■■■■ admin-app/                # SPA Admin Panel (Vue 3, Vite, TailwindCSS)
  ■   ■■■■ src/
  ■   ■   ■■■■ views/           # 11 Modul Tampilan Admin SPA
  ■   ■   ■■■■ stores/         # Pinia Auth & Global State Stores
  ■   ■   ■■■■ router/         # Vue Router Configuration
  ■   ■■■■ package.json
  ■■■■ src/                      # Backend Application Core (Node.js & Express)
  ■   ■■■■ config/             # Database (`database.js`), Env (`env.js`), WA Config
  ■   ■■■■ routes/             # Route Handlers (`admin.js`, `public.js`, `freelance-portal.js`)
  )
  ■   ■■■■ services/           # Business Logic Services (Drive Importer, OAuth, WA Service, Cron)
  ■   ■■■■ main.js              # Application Entry Point
  ■■■■ public/                  # Static Assets & Public Client HTML Pages
  ■   ■■■■ admin/               # Compiled SPA Production Assets
  ■   ■■■■ index.html           # Landing Page & Dynamic Portfolio
  ■   ■■■■ inquiry.html         # Public Booking Reservation Form
  ■   ■■■■ confirm-booking.html # Client Token Confirmation Page
  ■   ■■■■ freelance-portal.html # Simplified Freelance Portal
  ■   ■■■■ select-photos.html   # Client Selection Photo Gallery
  ■   ■■■■ tracking.html        # Order Progress Tracker Page
  ■   ■■■■ moodboard.html       # Inspiration Moodboard Collection Page
  ■■■■ DATA/                  # Storage Directory
  ■   ■■■■ wisuda.db            # SQLite Database File
  ■   ■■■■ uploads/             # Media & Documents Storage
  ■   ■■■■ backups/             # Automatic Database Backups
  ■■■■ package.json            # Root Dependencies & Scripts
```

Bab 2: Diagram Visual Arsitektur & Alur Kerja Frontend-Backend

2.1 Diagram Arsitektur Sistem (System Architecture Diagram)

flowchart TB

```

    subgraph Client_Layer ["Layer Antarmuka Frontend (Client & Admin)"]
        PublicPages["Public Pages\n(index.html, inquiry.html, tracking.html)"]
        FreelancePortal["Simplified Freelance Portal\n(freelance-portal.html)"]
        AdminSPA["Admin Panel SPA\n(Vue 3 + Pinia + Vite)"]
    end

    subgraph Backend_Layer ["Layer Backend Node.js / Express Core"]
        AuthMiddleware["Session Auth & Token Security"]
        AdminRoutes["Admin Routes Controller\n(/api/admin/*)"]
        PublicRoutes["Public Routes Controller\n(/api/public/*)"]
        FreelanceRoutes["Freelance Portal Routes\n(/api/public/freelance-portal/*)"]
        DriveImporter["Drive Importer & Retention Service"]
    end

    subgraph Database_Storage ["Layer Database & Filesystem"]
        SQLiteDB["SQLite Database\nDATA/wisuda.db (WAL Mode)"]
        LocalUploads["Local Uploads Storage\n(DATA/uploads/*)"]
    end

    subgraph External_Cloud ["Layanan External Cloud & WA"]
        GoogleDrive["Google Drive API v3\n(OAuth2 & Service Account)"]
        WhatsAppGateway["WhatsApp Service Gateway\n(Wa.me Direct Link)"]
    end

    PublicPages --> PublicRoutes
    FreelancePortal --> FreelanceRoutes
    AdminSPA --> AuthMiddleware --> AdminRoutes

    AdminRoutes --> SQLiteDB
    AdminRoutes --> LocalUploads
    PublicRoutes --> SQLiteDB
    FreelanceRoutes --> SQLiteDB

    DriveImporter --> GoogleDrive
    AdminRoutes --> GoogleDrive
    AdminRoutes --> WhatsAppGateway
    FreelanceRoutes --> WhatsAppGateway
  
```

2.2 Diagram Alur Urutan End-to-End (Sequence Diagram Workflow)

```
sequenceDiagram
    autonumber
    actor Client
    actor Admin
    actor Fotografer
    participant Backend as Node.js Backend
    participant DB as SQLite DB
    participant Drive as Google Drive

    rect rgb(240, 248, 255)
    note over Client, DB: Tahap 1: Inquiry & DP Payment
    Client->>Backend: Submit Form Reservasi (inquiry.html)
    Backend->>DB: INSERT INTO inquiries (status='new')
    Admin->>Backend: Kirim Quotation (InquiriesView.vue)
    Backend->>DB: INSERT INTO booking_tokens & UPDATE inquiries (status='quoted')
    Backend-->>Admin: Return Link WA Quote
    Client->>Backend: Bayar DP & Submit Token (confirm-booking.html)
    Admin->>Backend: Verifikasi DP
    Backend->>DB: INSERT INTO bookings (dp_status='paid')
    end

    rect rgb(255, 250, 240)
    note over Admin, Fotografer: Tahap 2: Assignment & Sesi Foto
    Admin->>Backend: Assign Fotografer (BookingsView.vue)
    Backend->>DB: INSERT INTO assignments (status='accepted')
    Fotografer->>Backend: Cek Jadwal (freelance-portal.html)
    Fotografer->>Backend: Klik 'Photo Shoot Selesai'
    Backend->>DB: UPDATE assignments (status='done')
    alt Payment 100% Lunas dari Awal
        Backend->>DB: UPDATE bookings (status='editing')
    else Pembayaran Masih DP 50%
        Admin->>Backend: Verifikasi Pelunasan (verify-balance)
        Backend->>DB: UPDATE bookings (balance_status='paid', status='editing')
    end
    end

    rect rgb(245, 255, 250)
    note over Admin, Drive: Tahap 3: Post Production 3-Langkah
    Admin->>Backend: Klik '■ Terima File' (DeliverablesView.vue)
    Backend->>DB: UPDATE deliverables & assignments (status='done')
    Admin->>Backend: Upload Drive Staging (upload-staging)
    Backend->>Drive: Scan Photos in Staging Folder
    Drive-->>Backend: Return Staged Photos Count
    Admin->>Backend: Klik '■ Push Staging' (bouncing button active)
    Backend->>DB: UPDATE bookings (selection_status='ready')
    Client->>Backend: Pilih Foto Favorit (select-photos.html)
    Admin->>Backend: Upload & Klik '■ Push Final Edit'
    Backend->>DB: UPDATE bookings (status='delivered', download_url)
    Backend-->>Client: Kirim Link Download Foto Final via WA
    end
```

2.3 Diagram Transisi State Machine Booking & Pasca Produksi

```
stateDiagram-v2
    [*] --> NewInquiry: Client Submit Form

    state "Tahap 1: Inquiry" as InquiryPhase {
        NewInquiry --> Quoted: Admin Kirim Quote Token
        Quoted --> DealtDP: Client Bayar DP 50% & Verifikasi
    }

    state "Tahap 2: Client & Booking" as BookingPhase {
        DealtDP --> Assigned: Admin Assign Fotografer
        Assigned --> ShootDone: Fotografer Klik 'Photo Shoot Selesai'
        ShootDone --> FullyPaid: Admin Verifikasi Pelunasan (100% Lunas)
    }

    state "Tahap 3: Post Production" as PostProductionPhase {
        FullyPaid --> MenungguFile: Masuk Post Production
        MenungguFile --> MenungguUploadStaging: Admin Klik '■ Terima File'
        MenungguUploadStaging --> ReadyPushStaging: Admin Upload Foto Staging Drive
        ReadyPushStaging --> MenungguPilihanClient: Admin Klik '■ Push Staging'
        MenungguPilihanClient --> EditHighlight: Client Submit Pilihan Foto
        EditHighlight --> HighlightReady: Admin Upload Foto Highlight
        HighlightReady --> FinalDelivered: Admin Klik '■ Push Final Edit'
    }

    FinalDelivered --> Completed: Client Terima Berkas Final
    Completed --> [*]
```

Bab 3: Spesifikasi Skema Database SQLite & Diagram ERD

3.1 Diagram ERD Relasi Database (Entity-Relationship Diagram)

```

erDiagram
    inquiries ||--o{ booking_tokens : generates
    inquiries ||--o{ bookings : converts_to
    packages ||--o{ bookings : applies_to
    bookings ||--o{ assignments : assigns
    freelancers ||--o{ assignments : assigned_to
    assignments ||--o{ deliverables : produces
    freelancers ||--o{ payouts : receives
    bookings ||--o{ portfolio_items : creates

    bookings {
        int id PK
        int inquiry_id FK
        int package_id FK
        string client_name
        string status
        string dp_status
        string balance_status
        string selection_status
        string tracking_token
    }

    assignments {
        int id PK
        int booking_id FK
        int fg_id FK
        string status
        int fg_fee
    }

    deliverables {
        int id PK
        int assignment_id FK
        int booking_id FK
        string delivery_type
    }

    inquiries {
        int id PK
        string client_name
        string status
        int transport_charge
    }

    freelancers {
        int id PK
        string name
        string access_code
        int active
    }

    payouts {
        int id PK
        int fg_id FK
        int amount
        string transfer_ref
    }

```

3.2 Detail Rinci 12 Tabel Database

Database SQLite (`DATA/wisuda.db`) berjalan menggunakan driver `better-sqlite3` dengan mode **Write-Ahead Logging (WAL)**.

1. Tabel `bookings`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `inquiry\_id` INTEGER (FK references `inquiries.id`)
- `package\_id` INTEGER (FK references `packages.id`)
- `client\_name` TEXT NOT NULL
- `client\_phone` TEXT NOT NULL
- `graduation\_date` DATE NOT NULL
- `shooting\_time` TEXT DEFAULT '09:00'
- `location` TEXT
- `university` TEXT
- `city` TEXT
- `total\_price` INTEGER DEFAULT 0
- `dp\_amount` INTEGER DEFAULT 0
- `dp\_status` TEXT DEFAULT 'pending'
- `dp\_verified\_by` INTEGER (FK references `users.id`)
- `dp\_verified\_at` DATETIME
- `dp\_bukti\_url` TEXT
- `balance\_amount` INTEGER DEFAULT 0
- `balance\_status` TEXT DEFAULT 'pending'
- `balance\_verified\_by` INTEGER (FK references `users.id`)
- `balance\_verified\_at` DATETIME
- `balance\_bukti\_url` TEXT
- `status` TEXT DEFAULT 'confirmed'
- `selection\_status` TEXT DEFAULT 'pending'
- `drive\_parent\_url` TEXT
- `staging\_drive\_url` TEXT
- `highlight\_drive\_url` TEXT
- `download\_url` TEXT
- `tracking\_token` TEXT UNIQUE
- `staged\_photo\_count` INTEGER DEFAULT 0
- `highlight\_photo\_count` INTEGER DEFAULT 0
- `final\_photo\_count` INTEGER DEFAULT 0

2. Tabel `assignments`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `booking\_id` INTEGER NOT NULL (FK references `bookings.id` ON DELETE CASCADE)
- `fg\_id` INTEGER NOT NULL (FK references `freelancers.id`)

- `status` TEXT DEFAULT 'accepted'
- `fg\_fee` INTEGER DEFAULT 0
- `offer\_status` TEXT DEFAULT 'accepted'
- `fg\_confirmed\_at` DATETIME
- `shoot\_start\_at` DATETIME
- `shoot\_end\_at` DATETIME
- `upload\_deadline` DATETIME

3. Tabel `deliverables`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `assignment\_id` INTEGER (FK references `assignments.id` ON DELETE CASCADE)
- `booking\_id` INTEGER (FK references `bookings.id`)
- `delivery\_type` TEXT DEFAULT 'link'
- `drive\_folder\_url` TEXT
- `raw\_folder\_url` TEXT
- `qc\_status` TEXT DEFAULT 'pending'
- `notes` TEXT

4. Tabel `inquiries`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `client\_name` TEXT NOT NULL
- `client\_phone` TEXT NOT NULL
- `graduation\_date` DATE NOT NULL
- `shooting\_time` TEXT DEFAULT '09:00'
- `location` TEXT
- `university` TEXT
- `city` TEXT
- `transport\_charge` INTEGER DEFAULT 0
- `status` TEXT DEFAULT 'new'

5. Tabel `freelancers`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `name` TEXT NOT NULL
- `phone` TEXT NOT NULL
- `city` TEXT
- `access\_code` TEXT UNIQUE
- `default\_rate` INTEGER DEFAULT 0
- `active` INTEGER DEFAULT 1

6. Tabel `packages`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `name` TEXT NOT NULL
- `price` INTEGER NOT NULL

- `max\_selected\_photos` INTEGER DEFAULT 15
- `highlight\_count` INTEGER DEFAULT 5
- `category` TEXT DEFAULT 'Standard'

7. Tabel `portfolio\_items`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `booking\_id` INTEGER (FK references `bookings.id`)
- `client\_initial` TEXT
- `university` TEXT
- `graduation\_year` INTEGER
- `cover\_image\_url` TEXT
- `published` INTEGER DEFAULT 1

8. Tabel `portfolio\_import\_jobs`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `booking\_id` INTEGER NOT NULL
- `folder\_id` TEXT NOT NULL
- `status` TEXT DEFAULT 'pending'
- `imported\_count` INTEGER DEFAULT 0

9. Tabel `booking\_tokens`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `inquiry\_id` INTEGER NOT NULL
- `token` TEXT UNIQUE NOT NULL
- `expires\_at` DATETIME NOT NULL
- `used` INTEGER DEFAULT 0

10. Tabel `payouts`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `fg\_id` INTEGER NOT NULL (FK references `freelancers.id`)
- `amount` INTEGER NOT NULL
- `transfer\_ref` TEXT UNIQUE NOT NULL
- `status` TEXT DEFAULT 'transferred'
- `created\_at` DATETIME DEFAULT CURRENT_TIMESTAMP

11. Tabel `fg\_schedules`

- `id` INTEGER PRIMARY KEY AUTOINCREMENT
- `fg\_id` INTEGER NOT NULL
- `date` DATE NOT NULL
- `status` TEXT DEFAULT 'available'

12. Tabel `settings`

- `key` TEXT PRIMARY KEY
- `value` TEXT

Bab 4: SOP Alur Operasional 3-Tahap Pasca Produksi (End-to-End Workflow)

4.1 Diagram Presisi Transformasi UI 3-Langkah Post Production

```
flowchart LR
    Langkah1["Langkah 1: Menunggu File\nBadge: Menunggu File / Berkas\nButton: [■ Terima File] (Amber)"]
    Langkah2["Langkah 2: Upload Staging\nBadge: Menunggu Upload Staging\nButton: [■ Upload File] (Drive)"]
    Langkah3["Langkah 3: Push Staging\nBadge: Ready Push (X File)\nButton: [■ Push Staging] (Animated Bounce Active)"]
    FinalState["Galeri Seleksi Terpublikasi ke Client"]

    Langkah1 -->|Admin Klik Terima File| Langkah2
    Langkah2 -->|Photos Uploaded > 0| Langkah3
    Langkah3 -->|Admin Klik Push Staging| FinalState
```

Bab 5: Panduan Integrasi Google Drive 3-Step Wizard

- Step 1 (Credentials Test):** Probe verification test ke `https://oauth2.googleapis.com/token`.
- Step 2 (OAuth Link):** Penautan akun Gmail Studio via OAuth2.
- Step 3 (Root Folder Structuring):** Struktur otomatis `Wisuda\_{client\_name}\_{year}`.

Bab 6: Penanganan Media, Background Worker & Kebijakan Retensi Storage

Direktori Storage	Jenis File	Aturan Retensi Cleanup
`DATA/uploads/portfolio/`	WebP Portfolio Published	Permanen (sampai dihapus admin)
`DATA/uploads/gallery_cache/`	Cache Thumbnail Proxy (`w400`/`w800`)	Dibersihkan otomatis saat highlight diupload atau TTL 7 hari.
`DATA/uploads/staging_uploads/`	Foto mentah sementara	Otomatis dihapus saat delivery final edit selesai.
`DATA/uploads/payment-proofs/`	Bukti Transfer DP / Pelunasan	Otomatis dihapus oleh Cron setelah 90 hari booking completed.
`DATA/uploads/invoices-client/`	PDF Kontrak & Invoice Client	Otomatis dihapus oleh Cron setelah 30 hari booking completed.

Bab 7: Panduan Seluruh Halaman Publik Client

- `index.html` — Landing Page & Portofolio Dinamis.
- `inquiry.html` — Form Reservasi Wisuda.
- `confirm-booking.html` — Konfirmasi Token Reservasi Client.

4. `select-photos.html` — Galeri Seleksi Foto Client.
5. `tracking.html` — Pelacakan Progres Order Real-Time.
6. `moodboard.html` — Koleksi Inspirasi Foto Client.
7. `invoice.html` — Invoice Digital & Proof PDF.

Bab 8: Matriks Lengkap Seluruh Tombol Aksi UI & Endpoint Backend

Halaman UI	Tombol / Aksi	Endpoint Backend API	Dampak Query SQL / Perubahan Data
InquiriesView	Kirim Quotation	`POST /api/admin/inquiries/:id/quote`	Update `inquiries.status = 'quoted'`, buat `booking_tokens`.
BookingsView	Assign FG	`POST /api/admin/bookings/:id/assign-fg`	Insert/Update `assignments` dengan `status = 'accepted'`.
BookingsView	Verifikasi DP	`POST /api/admin/bookings/:id/verify-dp`	Update `bookings.dp_status = 'paid'`, buat `bookings` record.
BookingsView	Verifikasi Pelunasan	`POST /api/admin/bookings/:id/verify-balance`	Update `bookings.balance_status = 'paid'`, jika shoot done → `status = 'editing'`.
DeliverablesView	**■ Terima File**	`POST /api/admin/post-production/:id/confirm-done`	Update `assignments.status = 'done'`, insert `deliverables.delivery_type = 'fisik'`, update `bookings.status = 'editing'`.
DeliverablesView	**■■ Upload File**	`POST /api/admin/post-production/:id/upload-staging`	Update `bookings.staging_drive_url`.
DeliverablesView	**■ Push Staging**	`POST /api/admin/post-production/:id/publish-staging`	Update `bookings.selection_status = 'ready'`.
DeliverablesView	**■ Push Highlight**	`POST /api/admin/post-production/:id/send-highlight-link`	Update `bookings.highlight_drive_url_unlocked`.
DeliverablesView	**■ Push Final Edit**	`POST /api/admin/post-production/:id/send-link`	Update `bookings.status = 'delivered'`, `download_url`.
PayrollView	Bayar Gaji	`POST /api/admin/payouts`	Insert `payouts` dengan `status = 'transferred'`.
Freelance Portal	■ Photo Shoot Selesai	`POST /api/public/freelance-portal/confirm-session`	Update `assignments.status = 'done'`, jika paid → `bookings.status = 'editing'`.
SettingsView	Verifikasi Google OAuth	`POST /api/admin/settings/drive-credentials`	Probe test ke Google Token API & insert ke `settings`.

Bab 9: Katalog Template Notifikasi WhatsApp & Otomatisasi Cron Jobs

9.1 Variabel Template WhatsApp

- `{client_name}` — Nama Client
- `{shooting_time}` — Jam Sesi Foto
- `{location}` — Lokasi Sesi Foto
- `{fg_name}` — Nama Fotografer
- `{tracking_url}` — Link Pelacakan Pesanan
- `{download_url}` — Link Drive Foto Final

9.2 Spesifikasi Cron Maintenance Jobs

- **Daily Capacity Cleanup (Tiap 00:00 WITA):** Pembersihan token kadaluarsa.
- **Drive Retention Cleanup (Tiap 02:00 WITA):** Pemindahan folder Drive expired ke Google Trash.
- **Database Auto Backup (Tiap 04:00 WITA):** Pembuatan cadangan database SQLite otomatis ke `DATA/backups/wisuda_auto_{timestamp}.db`.

Bab 10: Troubleshooting, Pemeliharaan & Developer Watermark

10.1 Troubleshooting Perbaikan Masalah Umum

1. Modal Error `Gagal mengonfirmasi file diterima`:

- *Solusi:* Backend menggunakan `UPDATE assignments SET status = 'done', shoot_end_at = CURRENT_TIMESTAMP WHERE id = ?`.

2. Tombol Push Staging Warna Abu-Abu (*Disabled*):

- *Solusi:* Tautkan folder Drive Staging yang berisi foto via tombol `Upload File`. Tombol Push akan otomatis bernyawa dengan animasi bergerak (*bounce*).

10.2 Developer Watermark (`AMS`)

Floating Developer Credit Bubble (`AMS`) terpasang di sudut kanan bawah setiap antarmuka web, mengintegrasikan versi commit Git dinamis.

Wisuda Platform Master Encyclopedic Manual v2.0 with Visual Mermaid Diagrams — Fully Verified.